

BRAC University (Department of Computer Science and Engineering)

CSE 221 (Algorithms) for Summer 2025 Semester

Assignment 02

Full Marks: 20

In this assignment, you will **simulate** the **Huffman Coding** to encode and decode a given text. The text to be encoded is your Student ID concatenated with your Full Name (with all letters converted to **UPPERCASE**).

Input Text:

If your Student ID is **19050122** and your Full Name is **Xyz Abcd Efghi**, then the text to be encoded becomes:

19050122-XYZ ABCD EFGHI

Rubric:

Frequency Calculation Count the frequency of each character in the text	4
Build Huffman Tree (Show all steps) Construct the Huffman Tree using the frequency table	6
Generate Huffman Codes Assign binary codes to each character based on the tree	3
Encoding Encode the first 10 characters in the text using the Huffman codes	3
Decoding - Decode the encoded binary string back to the original text. - Verify that the decoded text is exactly the same as the input text	4
Total Marks	20

SUBMISSION DEADLINE: 11:50 pm, 9 September, 2025

Instructions

- You must show all intermediate steps clearly in your submission.
- Start working early. If you submit at the last minute, there may be network issues.
- Once the submission portal is closed, late submission will be penalized.
- Penalty: Any late submission will result in deduction of 10 marks.